

DEEP NEURAL NETWORK

Course Code: AIMLCZG511

COMPLETE ANSWER BOOK

*Comprehensive Worked Solutions & Conceptual Derivations
for all 49 Practice Question Bank Problems*

Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division (WILP)

This solution book provides rigorous step-by-step mathematical calculations, backpropagation derivations, parameter counting, activation function comparisons, and diagnostic troubleshooting frameworks for practice questions.

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1 Module 1: Perceptrons, Linear Separability & Network Architecture

Question 1: Linear Inseparability in Single-Layer Perceptron

A bank uses a single-layer neural network to classify loan applications based on income and credit score. During testing, the classifier consistently misclassifies cases where the positive and negative examples cannot be separated by a straight line. Analyze the reason for this behavior and explain what architectural change would make the problem solvable.

Detailed Solution & Diagnostic Analysis:

- Mathematical Root Cause:** A single-layer Perceptron computes a decision boundary given by the linear equation:

$$z = w_1x_1 + w_2x_2 + b = 0 \quad (1)$$

where x_1 represents income, x_2 represents credit score, w_1, w_2 are input weights, and b is the bias. The output decision is produced via a threshold activation function $y = \mathbf{1}(z \geq 0)$. Consequently, a single-layer Perceptron can *only* form a hyperplane (a straight line in 2D space) as its decision boundary.

- Reason for Misclassification:** If loan approval features present non-linear interactions (e.g., high credit score compensating for low income, forming an XOR-like or circular decision boundary), positive and negative instances cannot be separated by any straight line. The single-layer network suffers from **high bias** (underfitting) and fails to converge to a zero-error decision boundary (Minsky & Papert theorem, 1969).
- Architectural Solution:** To resolve linear inseparability, the network must be upgraded to a **Multi-Layer Perceptron (MLP)** by inserting at least one **hidden layer** containing non-linear activation functions (such as ReLU, Sigmoid, or Tanh):
 - Feature Space Transformation:** Hidden neurons map the non-linearly separable 2D input space (x_1, x_2) into a higher-dimensional feature space $(h_1, h_2, \dots, h_k)$.
 - Non-linear Convex Polytopes:** Non-linear activations warp the coordinate space, allowing the output layer to draw a hyperplane that seamlessly separates the classes in the transformed feature space.

Key Concept & Exam Insight

Single-layer Perceptrons are limited strictly to linearly separable functions. Adding a hidden layer with non-linear activation functions enables universal function approximation, resolving complex feature interactions.

Question 2: Training Epochs vs. Architectural Limitations

A development team increases the number of epochs of a single-layer Perceptron because the model is unable to correctly classify a particular dataset. Explain whether increasing training iterations alone can overcome the limitation of the model.

Detailed Solution:

- 1. Perceptron Convergence Theorem:** According to Rosenblatt's Perceptron Convergence Theorem, if a training dataset is *linearly separable*, the Perceptron learning algorithm is guaranteed to converge to a correct weight vector in a finite number of steps (epochs).
- 2. Effect of Increasing Epochs on Linearly Inseparable Data:** If the dataset is non-linearly separable, increasing the number of training epochs **will never resolve the classification errors**. Instead, the weight vector will enter an infinite loop or oscillate endlessly near the boundary without minimizing the empirical risk to zero.
- 3. Conclusion:** Training iterations (epochs) optimize parameters within the hypothesis space defined by the model architecture. If the hypothesis space itself is incapable of representing non-linear boundaries, additional epochs add zero expressive capacity. The model requires an architectural upgrade (adding hidden layers and non-linearities), not additional training time.

Question 3: Single Neuron Forward Propagation Numerical

Consider a neuron with input vector $\mathbf{x} = [1, -2]$, weight vector $\mathbf{w} = [0.6, -0.4]$, and bias $b = 0.2$. Calculate the net input and output using an appropriate activation function.

Step-by-Step Numerical Calculation:

Step 1: Calculate Net Input (z):

$$z = \mathbf{w}^T \mathbf{x} + b = (w_1 \cdot x_1) + (w_2 \cdot x_2) + b \quad (2)$$

Substituting the given numerical values:

$$z = (0.6 \times 1) + (-0.4 \times -2) + 0.2 = 0.6 + 0.8 + 0.2 = 1.6 \quad (3)$$

Step 2: Calculate Output using Standard Sigmoid Activation $\sigma(z)$:

$$a = \sigma(z) = \frac{1}{1 + e^{-z}} = \frac{1}{1 + e^{-1.6}} \quad (4)$$

Evaluating $e^{-1.6} \approx 0.201897$:

$$a = \frac{1}{1 + 0.201897} = \frac{1}{1.201897} \approx \mathbf{0.8320} \quad (\text{or } 83.20\%) \quad (5)$$

Step 3: Alternative Output using ReLU Activation:

$$a_{\text{ReLU}} = \max(0, z) = \max(0, 1.6) = \mathbf{1.6} \quad (6)$$

Question 4: Role of Input, Hidden, and Output Layers

A classification system initially uses a Perceptron but later requires learning nonlinear decision boundaries. Compare the role of the input, hidden and output layers in the redesigned network.

Question 5: Depth and Hierarchical Feature Learning

A company replaces a shallow network with a deeper network for image recognition. Accuracy improves significantly, but the engineers are unable to explain why. Enable hierarchical feature learning using a suitable example.

Detailed Functional Comparison:

Layer Type	Primary Mathematical Role	Feature Representation	Activation Used
Input Layer	Passive buffer; receives raw features $\mathbf{x} \in \mathbb{R}^d$.	Represents raw feature space directly without transformation.	None (Identity)
Hidden Layer(s)	Performs non-linear feature transformation $\mathbf{h} = f(\mathbf{W}^{(1)}\mathbf{x} + \mathbf{b}^{(1)})$.	Warps non-linear space into linearly separable representations.	ReLU, Tanh, Sigmoid
Output Layer	Computes final class score/probability $\hat{\mathbf{y}} = g(\mathbf{W}^{(2)}\mathbf{h} + \mathbf{b}^{(2)})$.	Fits a linear decision hyperplane on the hidden features.	Sigmoid (Binary), Softmax (Multi-class), Linear (Regression)

Detailed Solution:

- Principle of Compositionality:** Deep networks express complex functions as a composition of simpler nested functions: $f(\mathbf{x}) = f_L(f_{L-1}(\dots f_1(\mathbf{x})))$.
- Hierarchical Feature Progression (Face Recognition Example):**
 - Early Layers (Layer 1–2):** Detect low-level primitive features such as oriented edges, color blobs, and fine spatial gradients.
 - Middle Layers (Layer 3–4):** Combine primitive edges into mid-level motifs and textures (e.g., curves, corners, eye contours, nose lines).
 - Late Layers (Layer 5+):** Aggregate mid-level motifs into high-level semantic object parts (e.g., full human faces, eyes, noses, mouths).
- Exponential Parameter Efficiency:** To represent the same complex function, a shallow network requires an exponentially larger number of hidden units (width $O(2^n)$), whereas a deep network achieves identical expressiveness with polynomial units ($O(n)$) across layers.

Question 6: Representation Transformation via Hidden Neurons

A medical-screening model performs well when the classes are linearly separable but fails when several interacting features determine the class. Explain how adding hidden neurons can transform the representation learned by the model.

Detailed Solution: When medical diagnostic features interact (e.g., age $\times$ blood pressure combined with genetic marker presence), the decision boundary in the raw feature space $\mathbb{R}^2$ is highly non-linear. Adding hidden neurons applies affine transformations followed by non-linear activations:

$$h_j = \sigma \left(\sum_i w_{ji} x_i + b_j \right) \quad (7)$$

Each hidden neuron acts as a custom non-linear decision boundary. Collectively, k hidden neurons project the input vector $\mathbf{x}$ into a k -dimensional latent space where interacting feature combinations map to distinct convex regions. The output layer then draws a flat hyperplane through this k -dimensional latent space, effortlessly separating high-risk from low-risk patient cases.

Question 7: Evaluating Neuron Count vs. Network Depth

A team argues that a sufficiently large number of neurons in a single-layer Perceptron should be able to solve any binary classification problem. Evaluate this claim using a suitable decision-boundary example.

Evaluation & Counter-Proof:

- 1. Evaluation of Claim:** The claim is **FALSE**.
- 2. Mathematical Proof via XOR Problem:** Consider the classic XOR dataset with inputs $(0, 0) \rightarrow 0$, $(1, 1) \rightarrow 0$, $(0, 1) \rightarrow 1$, and $(1, 0) \rightarrow 1$. A single-layer Perceptron with N neurons in parallel without hidden layering computes:

$$y = \text{act} \left(\sum_{i=1}^N w_i x_i + b \right) \quad (8)$$

Increasing the number of inputs or parallel weights does not alter the fundamental linear nature of the decision boundary $w_1 x_1 + w_2 x_2 + b = 0$. No matter how many parallel inputs or neurons are placed in a single layer, the overall network output remains a single linear hyperplane. It is impossible to separate $(0, 1)$ and $(1, 0)$ from $(0, 0)$ and $(1, 1)$ using a single line, proving that width without depth/hidden layering cannot solve non-linear problems.

2 Module 2: Activation Functions & Non-Linearities

Question 8: Vanishing Gradient with Sigmoid Units

A neural network uses Sigmoid activations in every hidden layer. As the network becomes deeper, the gradients reaching the first hidden layer become extremely small. Explain the phenomenon and recommend a more suitable activation strategy.

Detailed Mathematical Analysis:

- 1. Mathematical Root Cause:** The Sigmoid function is defined as $\sigma(z) = (1 + e^{-z})^{-1}$. Its first derivative with respect to z is:

$$\sigma'(z) = \sigma(z)(1 - \sigma(z)) \quad (9)$$

The maximum possible value of $\sigma'(z)$ occurs at $z = 0$, where $\sigma(0) = 0.5$, yielding:

$$\max \sigma'(z) = 0.5 \times (1 - 0.5) = \mathbf{0.25} \quad (10)$$

For $|z| > 2$, $\sigma'(z)$ rapidly decays toward zero (< 0.05).

- 2. Chain Rule Decay in Deep Networks:** During backpropagation through an L -layer network, the gradient of the loss with respect to the first layer weights $W^{(1)}$ involves a product of derivative terms:

$$\frac{\partial L}{\partial W^{(1)}} = \delta^{(L)} \cdot \prod_{l=2}^L \left(W^{(l)T} \cdot \sigma'(z^{(l)}) \right) \quad (11)$$

Since each $\sigma'(z^{(l)}) \leq 0.25$, multiplying L such terms causes the gradient to shrink exponentially as $(0.25)^L \rightarrow 0$. As a result, early layers receive virtually zero weight updates and fail to learn.

- 3. Recommended Activation Strategy:** Replace Sigmoid hidden units with **Rectified Linear Units (ReLU)** or **Leaky ReLU**:

$$f_{\text{ReLU}}(z) = \max(0, z), \quad f'_{\text{ReLU}}(z) = \begin{cases} 1 & \text{if } z > 0 \\ 0 & \text{if } z < 0 \end{cases} \quad (12)$$

Since the derivative is exactly 1.0 for all positive inputs, gradients pass through deep layers without exponential decay.

Question 9: Numerical Comparison of Sigmoid, ReLU, and Leaky ReLU

For a neuron with pre-activation $z = -2$, calculate the output of Sigmoid, ReLU, and Leaky ReLU with $\alpha = 0.01$. Compare their behavior.

Numerical Calculations & Comparison:

- 1. Sigmoid Activation $\sigma(-2)$:**

$$\sigma(-2) = \frac{1}{1 + e^{-(-2)}} = \frac{1}{1 + e^2} = \frac{1}{1 + 7.389056} = \frac{1}{8.389056} \approx \mathbf{0.1192} \quad (13)$$

- 2. ReLU Activation $\text{ReLU}(-2)$:**

$$\text{ReLU}(-2) = \max(0, -2) = \mathbf{0.0000} \quad (14)$$

- 3. Leaky ReLU Activation ($\alpha = 0.01$):**

$$\text{Leaky ReLU}(-2) = \max(0.01 \times (-2), -2) = -0.02 = \mathbf{-0.0200} \quad (15)$$

Behavioral Comparison:

- **Sigmoid:** Compresses output into range $(0, 1)$. Non-zero output (0.1192), but derivative is small (≈ 0.105), contributing to saturation.
- **ReLU:** Completely deactivates the neuron (output = 0). Derivative is exactly 0, blocking backpropagation entirely ("Dying ReLU").
- **Leaky ReLU:** Retains a small negative signal (-0.02) and a non-zero derivative ($\alpha = 0.01$), allowing gradients to flow back during training even when $z < 0$.

Question 10: Sigmoid Value, Derivative, and Gradient Propagation

For $z = 2$, calculate the Sigmoid activation and its derivative. Explain what the derivative indicates about gradient propagation.

Step-by-Step Calculation:

Step 1: Calculate Activation $a = \sigma(2)$:

$$a = \frac{1}{1 + e^{-2}} = \frac{1}{1 + 0.135335} = \frac{1}{1.135335} \approx \mathbf{0.8808} \quad (16)$$

Step 2: Calculate Derivative $\sigma'(2)$:

$$\sigma'(2) = \sigma(2) \cdot (1 - \sigma(2)) = 0.8808 \times (1 - 0.8808) = 0.8808 \times 0.1192 = \mathbf{0.10499} \approx \mathbf{0.1050} \quad (17)$$

Step 3: Interpretation for Gradient Propagation: The derivative value of 0.1050 implies that during backpropagation, any error gradient passing through this neuron is reduced to 10.5% of its incoming magnitude. If multiple such layers are stacked, $0.1050 \times 0.1050 \times \dots$ severely attenuates gradient signals, slowing optimization.

Question 11: Tanh Saturation and Slow Optimization

A deep network uses Tanh activations in several hidden layers. The training process becomes slow even though the loss decreases. Analyze whether Tanh could contribute to the problem.

Detailed Solution: Yes, Tanh contributes directly to training slowdown due to **gradient saturation**:

1. **Derivative Formula:** $\tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$, with derivative $\frac{d}{dz} \tanh(z) = 1 - \tanh^2(z)$.
2. **Saturation Regions:** When $|z| > 2.5$, $\tanh(z)$ approaches $+1$ or -1 , causing $1 - \tanh^2(z) \rightarrow 0$.
3. **Comparison with Sigmoid:** Tanh is **zero-centered** (range $[-1, 1]$), which avoids systematic zig-zag weight updates seen in Sigmoid. However, it still suffers from saturation in high-magnitude activation regimes, reducing weight gradients and causing slow training.

Question 12: Dying ReLU Problem and Technical Remedy

A computer-vision model replaces Sigmoid hidden units with ReLU. The model trains faster, but some neurons stop responding completely. Identify the possible problem and suggest a modification.

Diagnostic and Technical Remedy:

1. Problem Identification: This is the **"Dying ReLU" problem**. If a large gradient update causes a neuron's bias/weights to update such that $z = \mathbf{w}^T \mathbf{x} + b < 0$ for all training samples, ReLU outputs 0 with a gradient of 0. The neuron becomes permanently inactive and cannot recover.

2. Remedies:

- **Leaky ReLU:** $f(z) = \max(\alpha z, z)$ with $\alpha = 0.01$. Ensures non-zero gradient α for negative inputs.
- **Parametric ReLU (PReLU):** Learns α adaptively during backpropagation.
- **Exponential Linear Unit (ELU):** $f(z) = z$ for $z > 0$ and $\alpha(e^z - 1)$ for $z \leq 0$. Smoothly saturates to negative values.
- **Lower Learning Rate / He Initialization:** Prevents large weight updates that knock neurons into negative saturation.

Question 13: Output Layer Activation Choice for Unbounded Regression

A regression model requires outputs that can take both positive and negative values. Compare the suitability of ReLU, Sigmoid and Tanh for the output layer and justify a choice.

Activation	Output Range	Suitability for Unbounded Regression	Technical Justification
Sigmoid	$(0, 1)$	Unsuitable	Bounded; cannot predict negative values or values > 1 .
Tanh	$(-1, 1)$	Unsuitable	Bounded; cannot predict values outside $[-1, 1]$.
ReLU	$[0, \infty)$	Unsuitable	Truncates all negative predictions to zero.
Linear (Identity)	$(-\infty, \infty)$	Optimal / Recommended	Unbounded; passes raw $z = \mathbf{w}^T \mathbf{h} + b$ directly.

Question 14: ReLU vs. Leaky ReLU Dead Unit Issue

Two hidden-layer designs use ReLU and Leaky ReLU respectively. In the ReLU network, some units in the ReLU network permanently output zero. How does the alternative activation change the behavior?

Activation Suitability Analysis for Output Layer:

Detailed Solution: In standard ReLU, $f'(z) = 0$ for all $z \leq 0$. If a neuron's weights map all inputs to negative pre-activation values, gradient updates become:

$$\Delta w = -\eta \frac{\partial L}{\partial a} \cdot f'(z) \cdot x = -\eta \cdot \frac{\partial L}{\partial a} \cdot 0 \cdot x = 0 \quad (18)$$

The gradient vanishes entirely, trapping weights in their current state permanently. In **Leaky ReLU**, $f(z) = \alpha z$ for $z < 0$ (where $\alpha = 0.01$). The derivative for negative inputs is $f'(z) = \alpha \neq 0$. Consequently:

$$\Delta w = -\eta \cdot \frac{\partial L}{\partial a} \cdot \alpha \cdot x \neq 0 \quad (19)$$

This small gradient allows the optimization algorithm to update weights and nudge the neuron back into positive activation space.

3 Module 3: Forward/Backward Propagation, Loss Functions & Error Dynamics

Question 15: Predictive Maintenance Network Complete Forward Propagation

A predictive-maintenance network has two input neurons, two hidden Sigmoid neurons and one Sigmoid output neuron. Given the input vector $\mathbf{x} = [0.5, 0.8]$, hidden weights $W^{(1)} = \begin{bmatrix} 0.2 & 0.4 \\ -0.3 & 0.5 \end{bmatrix}$, hidden biases $\mathbf{b}^{(1)} = [0.1, -0.2]$, output weights $W^{(2)} = [0.7, -0.6]$, and output bias $b^{(2)} = 0.3$, perform complete forward propagation and calculate the final prediction.

Step-by-Step Mathematical Forward Propagation:

Step 1: Hidden Layer Pre-activations ($z_1^{(1)}, z_2^{(1)}$):

$$\begin{aligned} z_1^{(1)} &= w_{11}^{(1)}x_1 + w_{12}^{(1)}x_2 + b_1^{(1)} = (0.2 \times 0.5) + (0.4 \times 0.8) + 0.1 \\ &= 0.10 + 0.32 + 0.10 = \mathbf{0.52} \end{aligned} \quad (20)$$

$$\begin{aligned} z_2^{(1)} &= w_{21}^{(1)}x_1 + w_{22}^{(1)}x_2 + b_2^{(1)} = (-0.3 \times 0.5) + (0.5 \times 0.8) - 0.2 \\ &= -0.15 + 0.40 - 0.20 = \mathbf{0.05} \end{aligned} \quad (21)$$

Step 2: Hidden Layer Activations ($a_1^{(1)}, a_2^{(1)}$):

$$a_1^{(1)} = \sigma(0.52) = \frac{1}{1 + e^{-0.52}} = \frac{1}{1 + 0.5945205} \approx \mathbf{0.62715} \quad (22)$$

$$a_2^{(1)} = \sigma(0.05) = \frac{1}{1 + e^{-0.05}} = \frac{1}{1 + 0.951229} \approx \mathbf{0.51249} \quad (23)$$

Step 3: Output Layer Pre-activation ($z^{(2)}$):

$$\begin{aligned} z^{(2)} &= w_1^{(2)}a_1^{(1)} + w_2^{(2)}a_2^{(1)} + b^{(2)} \\ &= (0.7 \times 0.62715) + (-0.6 \times 0.51249) + 0.3 \\ &= 0.439005 - 0.307494 + 0.300000 = \mathbf{0.431511} \end{aligned} \quad (24)$$

Step 4: Final Prediction ($\hat{y} = a^{(2)}$):

$$\hat{y} = \sigma(0.431511) = \frac{1}{1 + e^{-0.431511}} = \frac{1}{1 + 0.649527} \approx \mathbf{0.6062} \quad (25)$$

Question 16: Mathematical Derivation of BCE Output Error Signal

For a binary classification network using Binary Cross-Entropy loss, derive the output-layer error term using the chain rule and show why it simplifies to $\delta^{(2)} = \hat{y} - y$.

Complete Mathematical Proof:

Step 1: Define Binary Cross-Entropy Loss (L) and Output Activation ($\hat{y}$):

$$L = -[y \ln(\hat{y}) + (1 - y) \ln(1 - \hat{y})], \quad \text{where } \hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}} \quad (26)$$

Step 2: Apply Chain Rule for Output Gradient $\delta = \frac{\partial L}{\partial z}$:

$$\delta = \frac{\partial L}{\partial z} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z} \quad (27)$$

Step 3: Compute First Partial Derivative $\frac{\partial L}{\partial \hat{y}}$:

$$\frac{\partial L}{\partial \hat{y}} = - \left(\frac{y}{\hat{y}} - \frac{1-y}{1-\hat{y}} \right) = \frac{-y(1-\hat{y}) + (1-y)\hat{y}}{\hat{y}(1-\hat{y})} = \frac{-y + y\hat{y} + \hat{y} - y\hat{y}}{\hat{y}(1-\hat{y})} = \frac{\hat{y} - y}{\hat{y}(1-\hat{y})} \quad (28)$$

Step 4: Compute Second Partial Derivative $\frac{\partial \hat{y}}{\partial z}$:

$$\frac{\partial \hat{y}}{\partial z} = \sigma'(z) = \hat{y}(1-\hat{y}) \quad (29)$$

Step 5: Multiply Term by Term:

$$\delta = \left(\frac{\hat{y} - y}{\hat{y}(1-\hat{y})} \right) \cdot [\hat{y}(1-\hat{y})] = \hat{y} - y \quad \blacksquare \quad (30)$$

Question 17: Binary Cross-Entropy Loss Calculation and Interpretation

A neural network predicts a probability of 0.25 for a sample whose actual class is 1. Calculate the Binary Cross-Entropy loss and interpret the result.

Calculation and Interpretation:

1. Formula: $L = -[y \ln(\hat{y}) + (1-y) \ln(1-\hat{y})]$. Given $y = 1, \hat{y} = 0.25$:

$$L = -1 \cdot \ln(0.25) - 0 = -\ln(1/4) = \ln(4) \approx \mathbf{1.3863} \quad (31)$$

2. Interpretation: A loss of 1.3863 is high (compared to $L \approx 0.105$ for $\hat{y} = 0.90$). Cross-Entropy penalizes confident incorrect predictions heavily. Because the model predicted a 75% chance of class 0 when actual label is 1, a large error signal is generated to drive swift weight adjustment.

Question 18: Negative Output Error Signal Interpretation

During backpropagation, the output neuron produces a negative error term. Explain what this indicates about the current prediction and the direction in which the output-layer weights are likely to move.

Detailed Solution:

1. Mathematical Sign Analysis: The error term is $\delta = \hat{y} - y$. If $\delta < 0$, it implies that $\hat{y} - y < 0 \implies \hat{y} < y$.

2. Prediction Status: The network is **underpredicting** the target value (e.g., predicting $\hat{y} = 0.2$ when target $y = 1.0$).

3. Weight Direction: The gradient with respect to output weight w_i is $\frac{\partial L}{\partial w_i} = \delta \cdot a_i^{(1)}$. Since $\delta < 0$ and hidden activations $a_i^{(1)} > 0$, the gradient is **negative**. Under Gradient Descent:

$$w_i^{\text{new}} = w_i^{\text{old}} - \eta \left(\frac{\partial L}{\partial w_i} \right) = w_i^{\text{old}} - \eta(\text{negative value}) = w_i^{\text{old}} + \eta|\text{gradient}| \quad (32)$$

Thus, the output-layer weights **increase**, which raises pre-activation z and drives predicted output $\hat{y}$ upwards toward target y .

Question 19: Chain Rule and Hidden Layer Gradient Errors

A student calculates the output-layer gradient correctly but obtains an incorrect hidden-layer gradient. Identify the role of the chain rule in propagating the error backward and explain where activation derivatives enter the calculation.

Detailed Solution: To compute the gradient for a hidden layer weight $w_{ji}^{(1)}$, the chain rule must backpropagate error from the output layer through the hidden activation:

$$\frac{\partial L}{\partial w_{ji}^{(1)}} = \frac{\partial L}{\partial z_j^{(1)}} \cdot \frac{\partial z_j^{(1)}}{\partial w_{ji}^{(1)}} = \delta_j^{(1)} \cdot a_i^{(0)} \quad (33)$$

The hidden error term $\delta_j^{(1)}$ is computed as:

$$\delta_j^{(1)} = \left(\sum_k w_{kj}^{(2)} \delta_k^{(2)} \right) \cdot f'(z_j^{(1)}) \quad (34)$$

Common student mistakes occur in two places:

- Omitting the hidden activation derivative $f'(z_j^{(1)})$.
- Evaluating f' using activation $a_j^{(1)}$ instead of pre-activation $z_j^{(1)}$ (e.g., using a instead of $1 - a^2$ for Tanh).

Question 20: Response to Confident Incorrect Predictions

A two-layer classifier gives a confident but incorrect prediction. Explain how the loss and output error signal would respond and why this can produce a comparatively large corrective update.

Detailed Solution: When a model gives a confident incorrect prediction (e.g., $y = 1$ but $\hat{y} = 0.001$):

- 1. Loss Magnitude:** BCE Loss $L = -\ln(0.001) \approx 6.91$ (extremely large).
- 2. Error Signal Magnitude:** Output error term $|\delta| = |\hat{y} - y| = |0.001 - 1.0| = 0.999 \approx 1.0$ (near maximum value).
- 3. Corrective Update:** Because $|\delta| \approx 1$, weight gradients $\frac{\partial L}{\partial w} = \delta \cdot h$ achieve maximum amplitude, forcing a large parameter update that quickly corrects the erroneous confidence.

Question 21: Impact of Changing Activation Functions

An engineer changes the activation function of a hidden layer but keeps the same weights and inputs. Explain which quantities in the forward pass and backpropagation would change and why.

Impacted Quantities:

- **Forward Pass Changes:**
 - Hidden activations $a^{(1)} = f(z^{(1)})$ change because functional mapping f changes.
 - Output pre-activations $z^{(2)} = W^{(2)}a^{(1)} + b^{(2)}$ and final predictions $\hat{y}$ change due to altered input features to the output layer.

- **Backpropagation Changes:**

- Output error signal $\delta^{(2)} = \hat{y} - y$ changes because $\hat{y}$ changed.
- Hidden layer error signal $\delta^{(1)} = (W^{(2)T}\delta^{(2)}) \odot f'(z^{(1)})$ changes due to both updated $\delta^{(2)}$ and the new derivative function f' .

4 Module 4: Optimization, Gradient Descent Variants & Gradient Pathology

Question 22: Parameter Update with Negative Gradient

A neural-network parameter has $w = 0.8$, learning rate $\eta = 0.1$, and gradient $g = -0.6$. Calculate the updated parameter and explain why the parameter increases despite gradient descent.

Calculation and Conceptual Explanation:

1. Gradient Descent Update Formula:

$$w^{\text{new}} = w^{\text{old}} - \eta \cdot g \quad (35)$$

Substituting $w = 0.8, \eta = 0.1, g = -0.6$:

$$w^{\text{new}} = 0.8 - (0.1 \times -0.6) = 0.8 - (-0.06) = \mathbf{0.86} \quad (36)$$

2. Why Parameter Increases: The gradient $g = \frac{\partial L}{\partial w}$ represents the slope of loss with respect to w . A negative gradient ($g < 0$) indicates that the loss function is **sloping downwards towards the right**. To decrease loss, the optimization algorithm must step in the opposite direction of the gradient (hence *gradient descent*), moving rightward and increasing w .

Question 23: Batch GD vs. SGD vs. Mini-Batch GD Parameter Updates

A training dataset produces sample gradients $g_1 = 0.4, g_2 = 0.6, g_3 = 0.2$, and $g_4 = 0.8$. Calculate the parameter update using Batch Gradient Descent, SGD using the first sample, and Mini-Batch GD using the first two samples.

Step-by-Step Comparative Calculations:

1. Batch Gradient Descent (Full Dataset, $N = 4$):

$$g_{\text{batch}} = \frac{1}{4} \sum_{i=1}^4 g_i = \frac{0.4 + 0.6 + 0.2 + 0.8}{4} = \frac{2.0}{4} = 0.50 \quad (37)$$

$$\Delta w_{\text{Batch}} = -\eta \cdot g_{\text{batch}} = \mathbf{-0.50 \eta} \quad (38)$$

2. Stochastic Gradient Descent (Sample 1 only):

$$g_{\text{SGD}} = g_1 = 0.40 \implies \Delta w_{\text{SGD}} = -\eta \cdot 0.40 = \mathbf{-0.40 \eta} \quad (39)$$

3. Mini-Batch Gradient Descent (First two samples, $B = 2$):

$$g_{\text{mini}} = \frac{g_1 + g_2}{2} = \frac{0.4 + 0.6}{2} = 0.50 \implies \Delta w_{\text{Mini}} = -\eta \cdot 0.50 = \mathbf{-0.50 \eta} \quad (40)$$

Question 24: Optimization Strategy Recommendation

A company trains a DNN using SGD and notices that the loss fluctuates heavily between iterations. Another implementation using Batch GD is extremely slow. Recommend a training strategy and justify your decision.

Recommendation and Justification: **Recommendation:** Implement **Mini-Batch Gradient Descent** with momentum or an adaptive optimizer (such as Adam).

- **Solves SGD Instability:** Averaging gradients across a batch (e.g., $B = 32, 64, 128$) reduces variance in gradient estimates, damping wild loss oscillations.
- **Solves Batch GD Slowness:** Mini-batches leverage modern GPU matrix parallel processing efficiently without requiring the entire dataset to fit into VRAM.

Question 25: Risks of Excessive Learning Rate Increases

A model's loss decreases very slowly after several epochs. The developer proposes increasing the learning rate by 20 times. Analyze the possible benefits and risks of this decision.

Trade-off Analysis:

- **Potential Benefit:** May escape flat plateaus or shallow local minima quickly if current learning rate is far too conservative.
- **Severe Risks (20× Increase):**
 - **Overshooting Minima:** Step size ηg becomes larger than the valley width, causing divergence where loss explodes to infinity or produces 'NaN' values.
 - **Oscillation:** The optimizer bounces back and forth across steep canyon walls without descending toward the minimum.

Question 26: Learning Rate Extremes: 0.001 vs 1.0

Two networks use learning rates of 0.001 and 1.0 respectively. Without performing numerical calculations, explain how their training behavior could differ.

Comparative Behavior:

- $\eta = 0.001$ (**Conservative Learning Rate**): Smooth, stable loss reduction; updates are small, minimizing risk of divergence. However, convergence is extremely slow and susceptible to getting stuck in local plateaus.
- $\eta = 1.0$ (**Aggressive Learning Rate**): Rapid early progress if initial gradients are small, but highly prone to severe loss oscillations, gradient explosion, and catastrophic divergence during training.

Question 27: Learning Rate Instability and Scheduling Strategy

A training run reaches a low loss initially but then becomes unstable immediately after the learning rate is increased. Diagnose the likely reason and suggest a more appropriate learning-rate strategy.

Diagnosis and Technical Solution:

1. **Diagnosis:** As optimization approaches a local minimum, the loss landscape bowl narrows. Increasing the learning rate causes step sizes to exceed the local basin radius, destabilizing updates and causing loss spikes.
2. **Recommended Strategy:** Use a **Learning Rate Scheduler** that reduces (decays) η over time:

- **Cosine Annealing:** Smoothly decays learning rate following a cosine curve.
- **ReduceOnPlateau:** Reduces η by factor of 0.1 when validation loss flattens.

Question 28: Full-Batch vs Mini-Batch Update Frequency and Convergence

Two teams use identical networks and datasets but obtain different convergence behavior because one uses full-batch updates and the other uses mini-batches. Explain how the update frequency and gradient estimates can account for the difference.

Aspect	Full-Batch Updates	Mini-Batch Updates
Gradient Nature	Exact, deterministic gradient of total loss surface.	Noisy, stochastic approximation of true gradient.
Update Frequency	Exactly 1 weight update per full epoch.	N/B weight updates per epoch (much higher frequency).
Saddle Point Escape	Can get trapped easily in flat saddle points.	Stochastic noise helps bounce out of shallow saddle points.

Question 29: Vanishing Gradient Multiplier Numerical

A 10-layer network receives an output gradient of 0.8. The gradient multiplier at every 0.25. Calculate the approximate gradient reaching the first layer.

Detailed Explanation:

Step-by-Step Calculation:

$$g_{\text{layer } 1} = g_{\text{output}} \times (m)^{10} = 0.8 \times (0.25)^{10} \quad (41)$$

Evaluating $(0.25)^{10} = (2^{-2})^{10} = 2^{-20} = \frac{1}{1,048,576} \approx 9.53674 \times 10^{-7}$:

$$g_{\text{layer } 1} = 0.8 \times 9.53674 \times 10^{-7} = \mathbf{7.6294 \times 10^{-7}} \quad (42)$$

Diagnosis: Classic **Vanishing Gradient**. The first layer gradient is virtually zero (0.000000763), halting early layer feature learning.

Question 30: Exploding Gradient Multiplier Numerical

Repeat the previous analysis when the gradient multiplier is 1.8. Identify the training problem.

Step-by-Step Calculation:

$$g_{\text{layer } 1} = 0.8 \times (1.8)^{10} \quad (43)$$

Evaluating $(1.8)^{10} \approx 357.046722$:

$$g_{\text{layer } 1} = 0.8 \times 357.046722 = \mathbf{285.6374} \quad (44)$$

Diagnosis: Severe **Exploding Gradient**. Gradient expands by over $350\times$, causing wild parameter updates and instability.

Question 31: Asymmetric Layer Learning Rate Pathology

An engineer observes that the final layers of a DNN learn rapidly while the first few layers hardly change. Diagnose the likely gradient problem and explain why it occurs.

Diagnosis and Explanation: This is caused by **Vanishing Gradients**. In backpropagation, output layers sit close to loss computation, receiving direct error signals $\delta^{(L)} = \hat{y} - y \approx O(1)$. Early layers sit behind multiple matrix multiplications and activation derivatives $\prod f'(z^{(l)})$. If activation derivatives are < 1 , gradient magnitudes decay exponentially with depth, starving front layers of updates while back layers update rapidly.

Question 32: Exploding Gradients and NaN Generation Remedies

A deep network suddenly produces extremely large parameter updates and eventually generates NaN values during training. Identify the likely cause and suggest two remedies.

Diagnosis and Remedies: **Cause: Exploding Gradients**, leading to numeric overflow ('inf'), which results in 'NaN' when combined in arithmetic operations. **Remedies:**

- 1. Gradient Clipping:** Rescales gradients if norm exceeds threshold: $\mathbf{g} \leftarrow \mathbf{g} \cdot \frac{\tau}{\|\mathbf{g}\|}$ if $\|\mathbf{g}\| > \tau$.
- 2. Batch Normalization / He Weight Initialization:** Standardizes layer activations, keeping gradient scale $O(1)$.

Question 33: Selection of Architectural Stabilization Techniques

A research group is choosing between Xavier initialization, He initialization, Batch Normalization, Residual Connections and Gradient Clipping. For a network suffering from unstable gradients, select two techniques and explain why they are appropriate.

Selection and Technical Justification:

Selection 1: Residual Connections (ResNets): Adds identity skip connections $y = f(x) + x$. During backpropagation, derivative $\frac{\partial y}{\partial x} = \frac{\partial f}{\partial x} + 1$. The $+1$ term acts as a gradient superhighway, allowing gradients to flow back to early layers without attenuation.

Selection 2: Batch Normalization: Normalizes mini-batch activations to zero mean and unit variance. Prevents internal covariate shift and keeps pre-activations out of saturating regions.

Question 34: Exponential Derivative Multiplication in Sigmoid Networks

A network using many Sigmoid layers shows almost no change in its early-layer weights. Explain how repeated multiplication of derivatives can create this behavior.

Detailed Explanation: Max value of Sigmoid derivative is 0.25. In an L -layer network, backpropagation multiplies $L - 1$ activation derivatives:

$$\text{Gradient Scale} \leq (0.25)^{L-1} \quad (45)$$

For $L = 5$, $(0.25)^4 = 0.0039$ (0.39% of signal). For $L = 10$, $(0.25)^9 = 0.0000038$ (0.00038%). Early layers receive virtually zero weight updates.

Question 35: Gradient Clipping vs Careful Initialization

A deep network's gradients become very large only for a small number of training batches. Explain why gradient clipping could be useful and distinguish its purpose from that of careful weight initialization.

Detailed Comparison:

- **Gradient Clipping:** A dynamic runtime safety guard that caps outlier gradient vectors during specific batch updates where loss surface exhibits extreme steepness/cliffs.
- **Careful Weight Initialization (Xavier/He):** A static setup procedure that sets variance of weights at step 0 so variance of activations remains constant across layers. Initialization cannot prevent localized loss cliffs during training; clipping actively handles them.

5 Module 5: Convolutional Neural Networks (CNNs)

Question 36: CNN Dimension Calculation after Conv and Max Pooling

A CNN receives a $28 \times 28 \times 1$ image. It contains a 3×3 convolution with 8 filters, stride 1 and no padding, followed by 2×2 max pooling with stride 2. Calculate the output dimensions after both layers.

Step-by-Step Spatial Dimension Calculation:

Layer 1: Convolutional Layer ($K = 3, S = 1, P = 0, N_f = 8$):

$$W_{\text{out}} = \left\lfloor \frac{W_{\text{in}} - K + 2P}{S} \right\rfloor + 1 = \left\lfloor \frac{28 - 3 + 0}{1} \right\rfloor + 1 = 25 + 1 = \mathbf{26} \quad (46)$$

Dimensions after Conv layer: $\mathbf{26 \times 26 \times 8}$.

Layer 2: Max Pooling Layer ($K_p = 2, S_p = 2$):

$$W_{\text{pool}} = \left\lfloor \frac{26 - 2}{2} \right\rfloor + 1 = 12 + 1 = \mathbf{13} \quad (47)$$

Final Output Dimensions: $\mathbf{13 \times 13 \times 8}$.

Question 37: RGB Conv Output Size and Trainable Parameter Count

A CNN receives an RGB image of size $64 \times 64 \times 3$. A convolution layer contains 16 filters of size 3×3 , stride 1 and padding 1. Calculate its output size and number of trainable parameters.

Step-by-Step Calculation:

Step 1: Output Spatial Dimensions ($W_{\text{in}} = 64, K = 3, S = 1, P = 1, N_f = 16$):

$$W_{\text{out}} = \left\lfloor \frac{64 - 3 + 2(1)}{1} \right\rfloor + 1 = 63 + 1 = \mathbf{64} \quad (48)$$

Output Size: $\mathbf{64 \times 64 \times 16}$.

Step 2: Trainable Parameter Count: Each filter has shape $K \times K \times C_{\text{in}} = 3 \times 3 \times 3 = 27$ weights, plus 1 bias. Total parameters for 16 filters:

$$\text{Params} = N_f \times (K \times K \times C_{\text{in}} + 1) = 16 \times (3 \times 3 \times 3 + 1) = 16 \times 28 = \mathbf{448} \quad (49)$$

Question 38: Parameter Count with Multiple Input Channels

A CNN has a convolution layer with 32 filters of size 5×5 , receiving 16 input channels. Calculate the number of trainable parameters including biases.

Step-by-Step Calculation:

$$\text{Params} = N_f \times (K_h \times K_w \times C_{\text{in}} + 1) \quad (50)$$

Given $N_f = 32, K_h = 5, K_w = 5, C_{\text{in}} = 16$:

$$\text{Weights per filter} = 5 \times 5 \times 16 = 400$$

$$\text{Weights} + \text{Bias per filter} = 400 + 1 = 401$$

$$\text{Total Trainable Parameters} = 32 \times 401 = \mathbf{12,832} \quad (51)$$

Question 39: Evaluating Parameter Claims for Pooling Layers

A student claims that pooling layers increase the number of trainable parameters because they introduce additional operations. Evaluate the claim.

Evaluation: The claim is **FALSE**. Pooling layers (Max Pooling or Average Pooling) perform fixed, non-parametric mathematical operations over sliding local windows. They contain **zero learnable parameters** (0 weights, 0 biases).

Question 40: Flattening Raw Pixels vs Convolutional Feature Extraction

A surveillance company considers two models: one that directly flattens a $128 \times 128 \times 3$ image and another that first applies convolution and pooling. Explain why the second approach is generally more suitable for image analysis.

Detailed Comparison:

- 1. Parameter Explosion in FC Network:** $128 \times 128 \times 3 = 49,152$ pixels. Connecting directly to a 1000-unit hidden layer requires $49,152 \times 1000 \approx 49.15$ million parameters in layer 1 alone, leading to severe overfitting.
- 2. Loss of 2D Spatial Structure:** Flattening destroys 2D spatial relationships and local pixel adjacencies.
- 3. ConvNet Advantages:** Parameter sharing, translation invariance, and local receptive fields reduce parameters dramatically while preserving 2D spatial features.

Question 41: Strided Convolution Output Dimensions and Resolution Trade-off

A CNN receives a $32 \times 32 \times 3$ image and applies a 3×3 convolution with stride 2 and padding 1. Determine the output dimensions and explain how changing the stride affects spatial resolution.

Calculation and Resolution Trade-off:

$$W_{\text{out}} = \left\lfloor \frac{32 - 3 + 2(1)}{2} \right\rfloor + 1 = \left\lfloor \frac{31}{2} \right\rfloor + 1 = 15 + 1 = \mathbf{16} \quad (52)$$

Output Dimensions: $\mathbf{16 \times 16 \times N_f}$. **Effect of Stride:** Increasing stride downsamples feature maps rapidly, reducing memory/compute, but can lose fine spatial detail.

Question 42: Impact of Doubling Filter Count

A model designer doubles the number of filters in a convolutional layer while keeping the kernel size unchanged. Explain which dimensions of the parameter count and feature representation are affected.

Affected Dimensions:

- **Parameter Count:** Exactly **doubles** (linear scaling with N_f).
- **Feature Representation:** Output channel depth C_{out} doubles, expanding expressive capacity to extract a wider diversity of visual patterns.
- **Spatial Dimensions ($H \times W$):** Unchanged.

Question 43: Receptive Field Calculation Layer-by-Layer

A CNN contains a sequence of 3×3 convolutions and 2×2 pooling layers. Determine the receptive field of a neuron in a later feature map with respect to the original image. Clearly show the calculation layer by layer.

Layer-by-Layer Receptive Field Calculation: Formula: $RF_l = RF_{l-1} + (K_l - 1) \cdot J_{l-1}$, where $J_{l-1} = \prod_{i=1}^{l-1} S_i$ is cumulative stride.

Layer (l)	Kernel (K_l)	Stride (S_l)	Cumulative Stride (J_l)	Receptive Field (RF_l)
Input Layer	—	—	1	1×1
Conv 1	3×3	1	1	$1 + (3 - 1) \times 1 = 3 \times 3$
Pool 1	2×2	2	2	$3 + (2 - 1) \times 1 = 4 \times 4$
Conv 2	3×3	1	2	$4 + (3 - 1) \times 2 = 8 \times 8$

Question 44: One 7×7 vs Three 3×3 Convolutions

A CNN designer replaces one 7×7 convolution with three consecutive 3×3 convolutions. Analyze the effect on receptive field, parameters and number of nonlinear transformations.

Detailed Comparative Analysis (C channels):

1. **Receptive Field:** Three stacked 3×3 convs give $1 + (3 - 1) + (3 - 1) + (3 - 1) = 7 \times 7$ effective receptive field, matching the single 7×7 layer.
2. **Parameter Efficiency:**
 - Single 7×7 : $1 \times (7 \times 7 \times C^2) = 49 C^2$
 - Three 3×3 : $3 \times (3 \times 3 \times C^2) = 27 C^2$ (a 45% reduction in parameters!).
3. **Non-linearities:** Three non-linear activation functions instead of one, giving greater discriminative capability.

Question 45: LeNet-5 vs. VGG Expressiveness

A company chooses between LeNet-5 and VGG-style architecture for a complex image-classification problem. Explain why the deeper architecture may learn richer representations.

Detailed Solution: VGG stacks deep sequences of small 3×3 filters with deep channel depth (512), creating a high-capacity feature hierarchy that extracts intricate visual semantics compared to shallow architectures like LeNet-5.

Question 46: Spatial Dimension Reduction in VGG Architecture

A VGG-style CNN receives a $224 \times 224 \times 3$ image and applies three 2×2 max-pooling operations with stride 2. Determine the spatial dimensions before the fully connected layer.

Sequential Downsampling Calculation:

Step 1: Input: 224×224 .

Step 2: Max Pool 1 ($S = 2$): $\frac{224}{2} = 112 \times 112$.

Step 3: Max Pool 2 ($S = 2$): $\frac{112}{2} = 56 \times 56$.

Step 4: Max Pool 3 ($S = 2$): $\frac{56}{2} = 28 \times 28$.

Spatial dimensions before FC layer: 28×28 .

Question 47: Autonomous Vehicle Vision Feature Progression

An autonomous-vehicle vision system requires detection of increasingly complex features—from edges to textures to object parts. Explain how a CNN's hierarchical feature extraction enables this progression. Compare this with directly connecting the pixels to a fully connected layer.

Detailed Solution: ConvNets extract visual representations hierarchically (edges $\rightarrow$ textures $\rightarrow$ object parts $\rightarrow$ vehicles/pedestrians) through local receptive fields and weight sharing. Fully connected networks lack spatial locality, requiring prohibitive parameter scale and failing to generalize across spatial translations.

Question 48: Trade-off of Early Convolution Stride

A CNN designer observes that increasing the stride of an early convolution reduces the size of later feature maps. Explain the trade-off between computational efficiency, spatial detail and receptive field.

Trade-off Analysis:

- **Higher Stride ($S \geq 2$):** Reduces memory and FLOPs dramatically, and expands receptive field faster.
- **Trade-off / Drawback:** Discards fine-grained spatial information prematurely, impairing small object detection.

Question 49: Small Defect Detection Failure and Architectural Fix

An image-classification model performs well on large objects but poorly on small defects. Analyze how pooling, stride and receptive-field growth could contribute to this behavior and suggest an architectural adjustment.

Analysis and Architectural Fix:

1. Cause: Aggressive pooling and high strides downsample spatial resolution too quickly, washing out sub-pixel or small-region defects.

2. Architectural Adjustments:

- Remove aggressive early pooling; reduce stride in initial conv layers.
- Use **Feature Pyramid Networks (FPN)** or **Skip Connections** (U-Net style) to merge high-resolution early features with late semantic features.
- Use **Dilated/Atrous Convolutions** to expand receptive fields without losing spatial resolution.